

Document Name: Master Batch Record: Production Bioreactor and Triple Transfection			Page 1	
Number: MBR-USP-201	Version: 1.0	Effective Date: 01APR2026	Status: Effective	Part No: DS-OBX-201

Record Reproduction Certification

This document is an accurate reproduction of the approved Master Manufacturing Record executed for this lot. Handwritten entries are shown in blue; preprinted form text is black.

Product and Lot Information

Product	OBX-201 (recombinant AAV9)	Drug substance lot	DS-26051
Process step	Production bioreactor, triple transfection	Manufacturing date	28APR2026
Cell line	Suspension HEK293	Working cell bank lot	WCB-293-018
Production bioreactor ID	BRX-200-02	Working volume (mL)	10,000
Transfection reagent	Linear PEI (1 mg/mL)	Target harvest	72 h post-transfection

Referenced SOPs

SOP Number	Title
SOP-3010	Working Cell Bank Thaw and Seed Train Expansion
SOP-3040	N-1 Seed Bioreactor Operation
SOP-3110	Triple Transfection and Production Culture
SOP-3120	Production Bioreactor In-Process Monitoring
SOP-3150	Harvest and Lysis
SOP-1203	Buffer and Solution Preparation
SOP-0905	Bioburden and Endotoxin Sampling

1. EQUIPMENT / INSTRUMENTS			
Description	EIN	Calibration Due	Performed By / Date
Production bioreactor skid	BRX-200-02	30SEP2026	TN 28APR26
Dissolved oxygen probe	DO-207	12OCT2026	TN 28APR26
pH probe	PHM-209	05NOV2026	TN 28APR26
Viable cell counter	VCC-011	20AUG2026	TN 28APR26
Analytical balance	BAL-021	22NOV2026	TN 28APR26
Biosafety cabinet	BSC-06	15DEC2026	TN 28APR26

2. PLASMID AND REAGENT PREPARATION

Component	Identity / Composition	Lot Number	Prepared By / Date	Verified By / Date
Rep/Cap plasmid	pOBX-RepCap (AAV9)	PL-REPCAP-0417	TN 28APR26	PA 28APR26
Helper plasmid	pHelper (Ad E2A, E4, VA RNA)	PL-HELP-0417	TN 28APR26	PA 28APR26
Transgene plasmid	pOBX-ITR-transgene	PL-TRANS-0417	TN 28APR26	PA 28APR26
Transfection reagent	Linear PEI, 1 mg/mL	PEI-0392	TN 28APR26	PA 28APR26
Production medium	Chemically defined, serum-free	MED-2208	TN 28APR26	PA 28APR26

3. SEED AND INOCULATION

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
3.1	Record viable cell density at inoculation.	Viable cell density (e6 cells/mL). Acceptance: 0.5 to 1.5.	0.8	TN 28APR26	PA 28APR26
3.2	Record viability at inoculation.	Viability (percent). Acceptance: not less than 90.	96	TN 28APR26	PA 28APR26
3.3	Confirm working volume.	Working volume (mL). Acceptance: 10,000 plus or minus 500.	10,000	TN 28APR26	PA 28APR26

4. TRANSFECTION

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
4.1	Record viable cell density at transfection.	Viable cell density (e6 cells/mL). Acceptance: 1.0 to 2.0.	1.6	TN 28APR26	PA 28APR26
4.2	Record culture volume at transfection.	Culture volume (mL). Acceptance: record.	10,000	TN 28APR26	PA 28APR26
4.3	Calculate total plasmid DNA for the transfection.	Total DNA (mg). Acceptance: 1.0 microgram/mL per SOP-3110. Formula: Total	DNA per mL (microgram/mL): 1.0 Culture volume (mL): 10,000	TN 28APR26	PA 28APR26

		DNA (mg) = DNA per mL (microgram/mL) x Culture volume (mL) / 1000	Result (mg): 10.0		
4.4	Calculate PEI mass from the DNA to PEI ratio.	PEI mass (mg). Acceptance: DNA to PEI 1:2 by weight per SOP-3110. Formula: PEI (mg) = Total DNA (mg) x 2	Total DNA (mg): 10.0 DNA:PEI ratio: 1:2 Result (mg): 20.0	TN 28APR26	PA 28APR26
4.5	Split plasmid DNA by molar ratio.	Rep/Cap : Helper : Transgene = 1:1:1 (mg each).	3.33 / 3.33 / 3.33	TN 28APR26	PA 28APR26
4.6	Record DNA-PEI complexation time before addition.	Complexation time (min). Acceptance: 15 to 30 min.	18	TN 28APR26	PA 28APR26
4.7	Confirm addition of transfection complex to the reactor.	Complex added to reactor?	☒ X Yes ☐ No	TN 28APR26	PA 28APR26

5. PRODUCTION CULTURE					
Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
5.1	Record temperature setpoint.	Temperature (C). Acceptance: 37.0 plus or minus 0.5.	37.0	TN 28APR26	PA 28APR26
5.2	Record dissolved oxygen setpoint.	Dissolved oxygen (percent). Acceptance: 40 plus or minus 10.	40	TN 28APR26	PA 28APR26
5.3	Record hour 24 viability.	Viability (percent). Acceptance: record.	97	TN 28APR26	PA 28APR26
5.4	Record hour 24 culture pH.	pH. Acceptance: 7.0 to 7.2.	7.08	TN 28APR26	PA 28APR26
5.5	Record hour 48 viability.	Viability (percent). Acceptance: record.	94	TN 28APR26	PA 28APR26
5.6	Record hour 48 culture pH.	pH. Acceptance: 7.0 to 7.2.	7.06	TN 28APR26	PA 28APR26
5.7	Record hour 72	Viability (percent).	88	TN 28APR26	PA 28APR26

	viability.	Acceptance: record.			
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6. HARVEST					
Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
6.1	Record harvest time post-transfection.	Harvest time (h). Acceptance: 48 to 96, target 72.	72	TN 01MAY26	PA 01MAY26
6.2	Record harvest viability.	Viability (percent). Acceptance: not less than 70.	82	TN 01MAY26	PA 01MAY26
6.3	Record harvest working volume.	Volume (mL). Acceptance: record.	9,900	TN 01MAY26	PA 01MAY26

7. IN-PROCESS SAMPLING					
Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
7.1	Record in-process vector genome titer (qPCR).	VG titer (e10 vg/mL). Acceptance: record.	9.8	TN 01MAY26	PA 01MAY26
7.2	Confirm bioburden sample collected.	Bioburden sample collected?	☒ X Yes ☐ No	TN 01MAY26	PA 01MAY26
7.3	Confirm mycoplasma sample collected.	Mycoplasma sample collected?	☒ X Yes ☐ No	TN 01MAY26	PA 01MAY26

8. PROCESS COMMENTS			
Page / Step	Comment	Performed By / Date	Verified By / Date
Step 5.6	Hour 48 pH initially transcribed as 7.60; single-line corrected to 7.06 against the instrument log. No process impact.	TN 28APR26	PA 28APR26

BATCH RECORD APPROVALS	
Manufacturing Review	Quality Assurance Review
F. Castellano 05JUN2026	L. Marchetti 05JUN2026

Manufacturing and QA review signatures are applied at batch disposition.